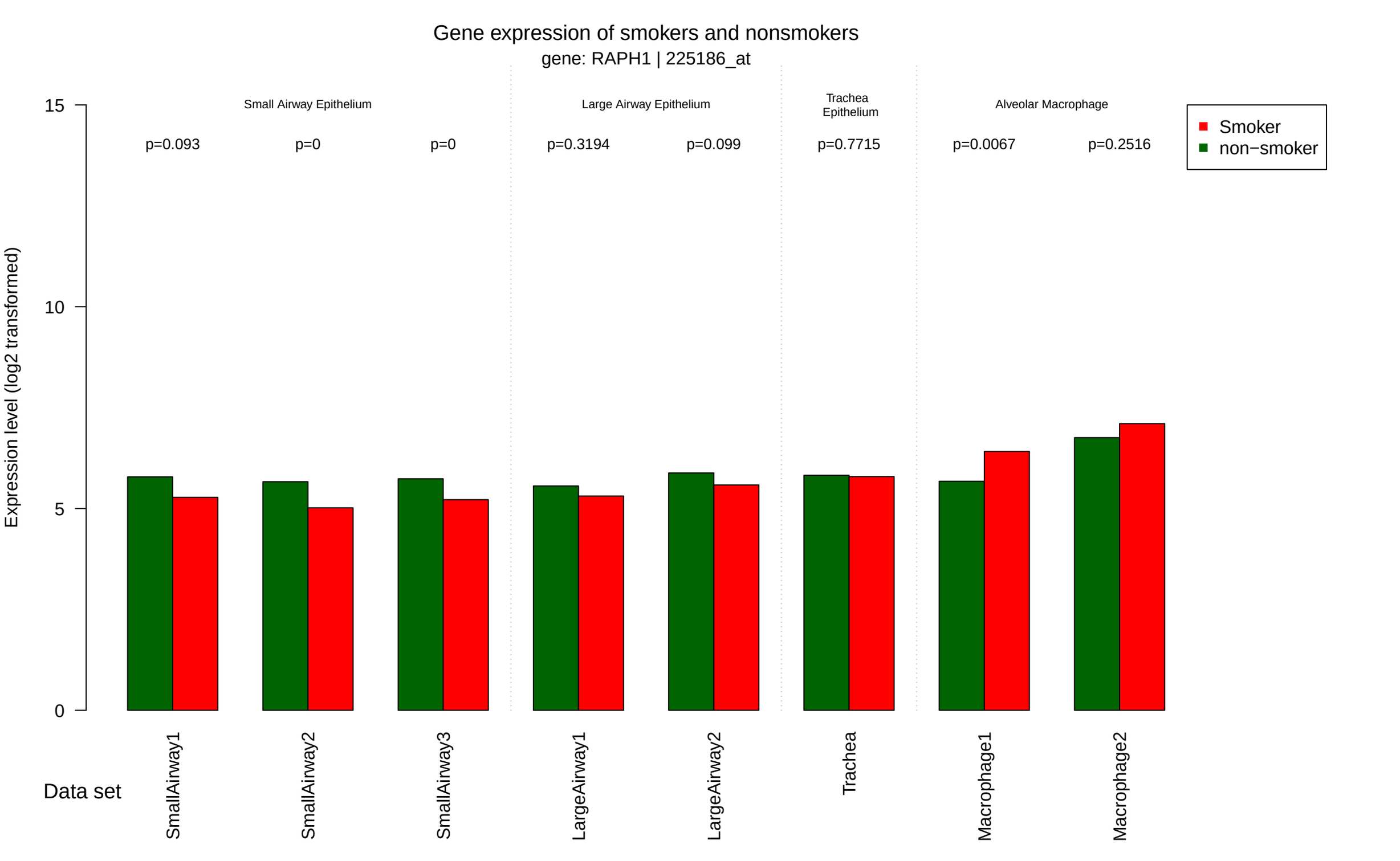
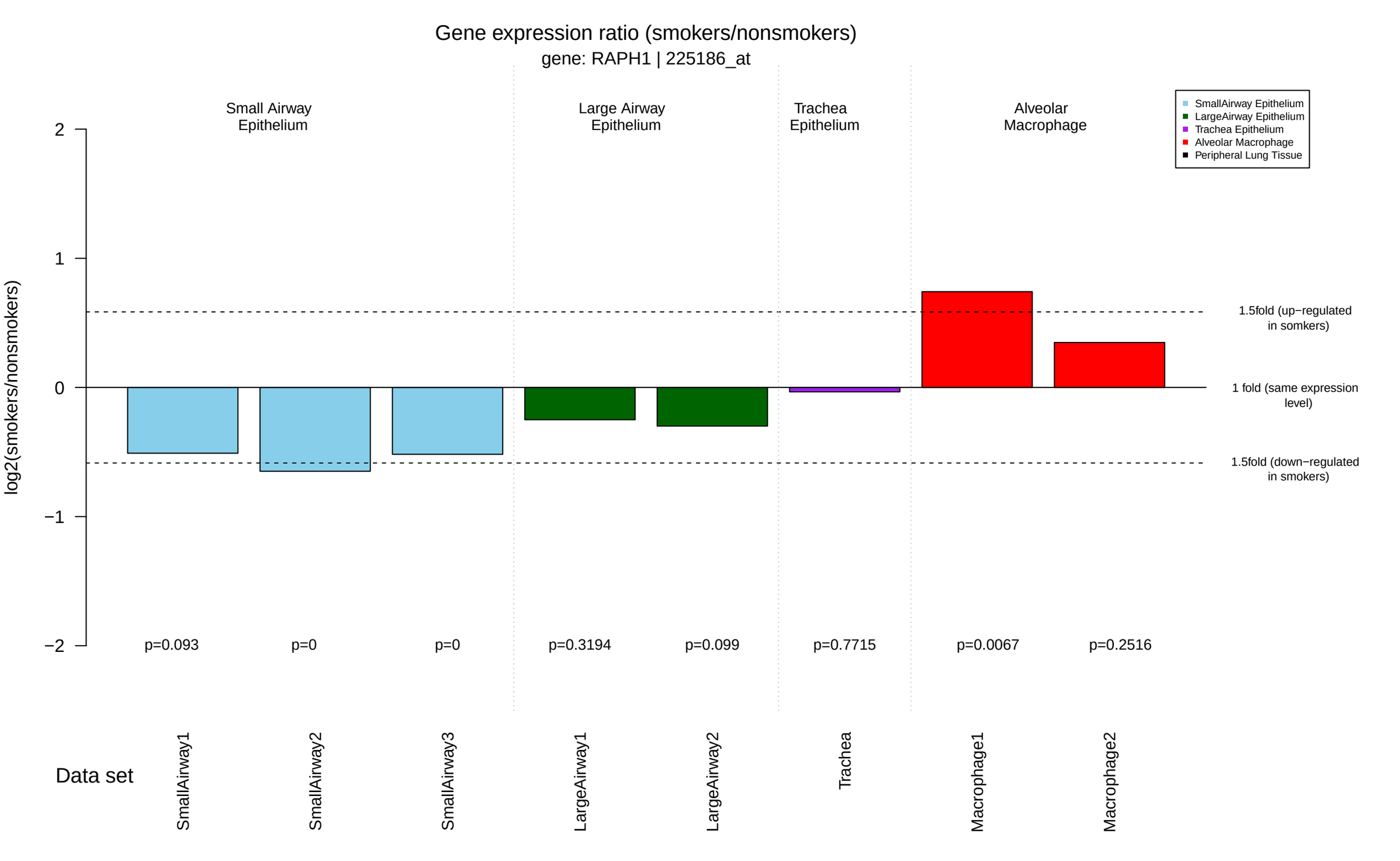
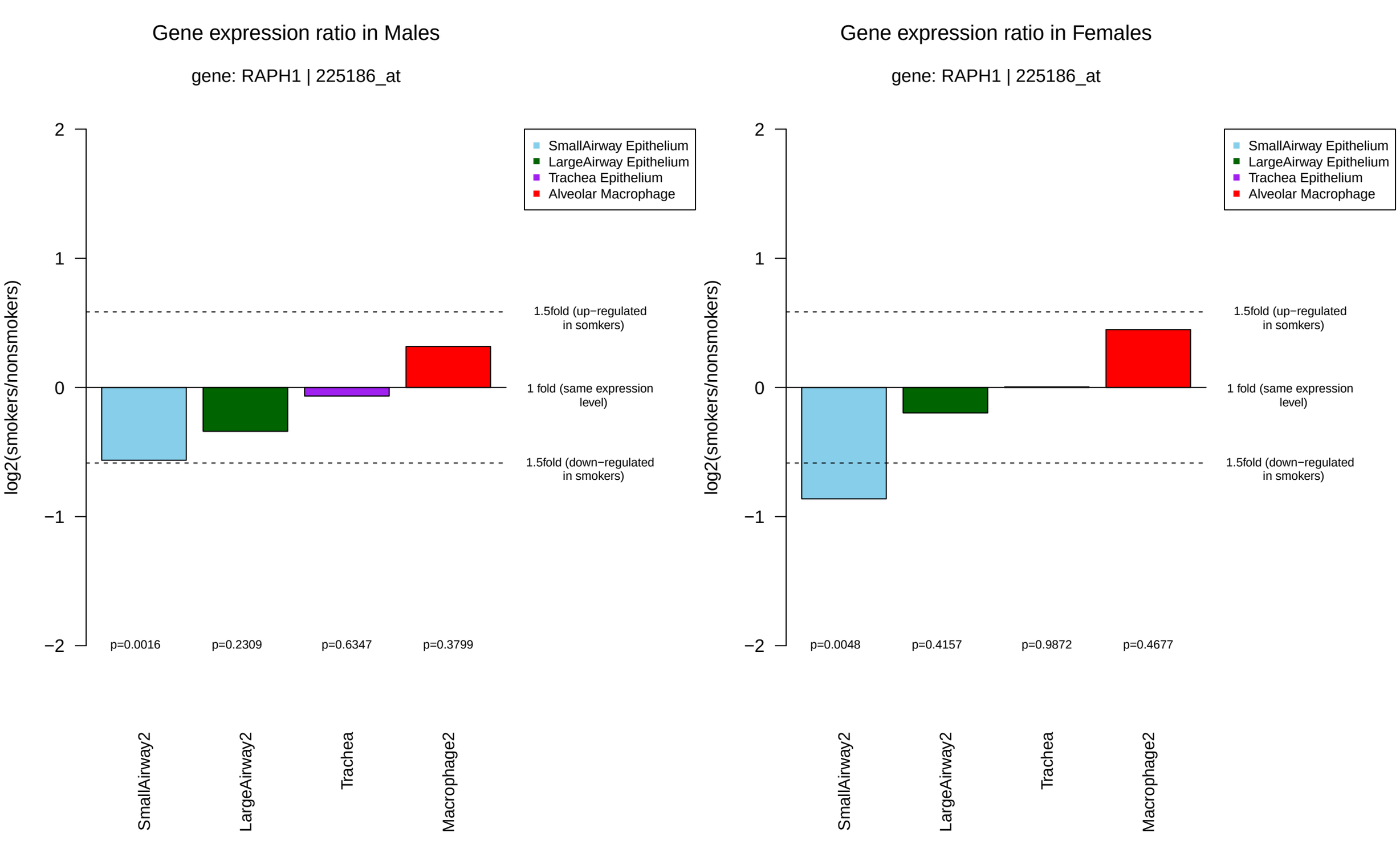
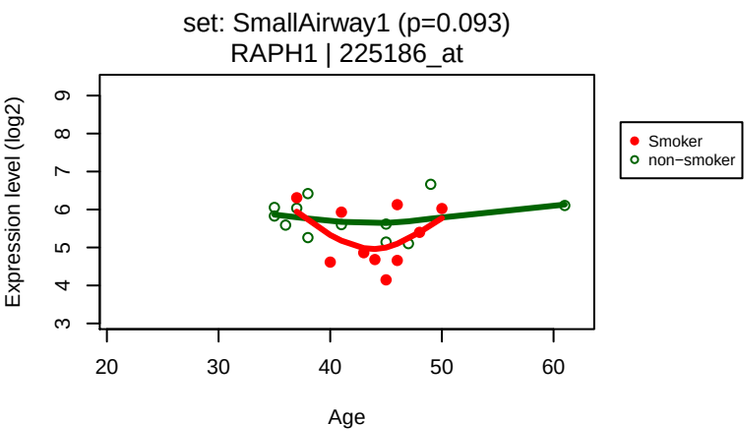
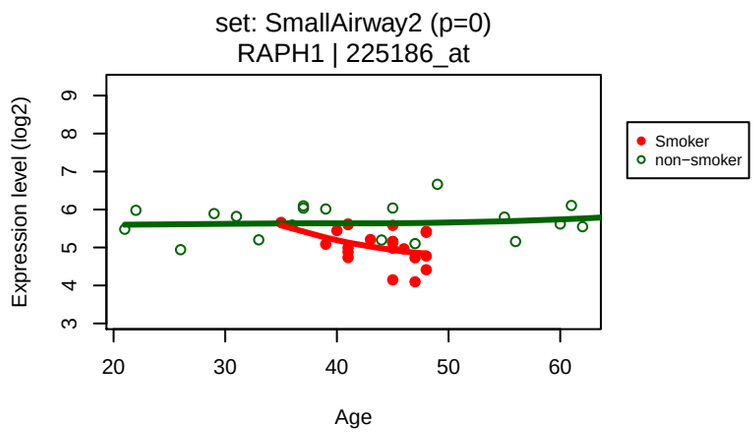
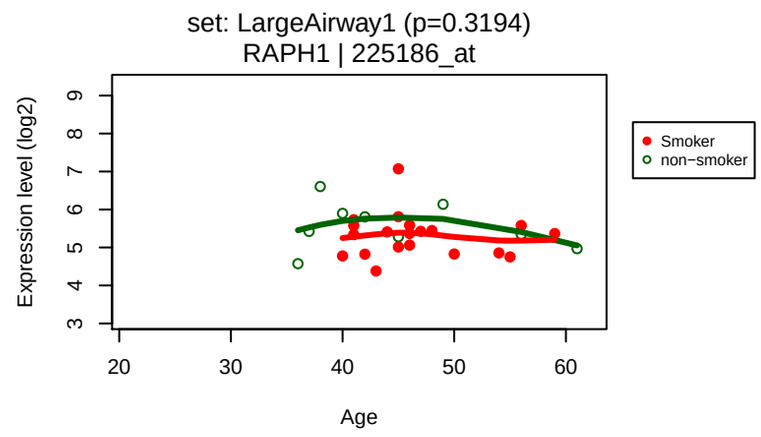

Gene expression of smokers and nonsmokers
gene: RAPH1 | 225186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.093
p=0
p=0
p=0.3194
p=0.099
p=0.7715
p=0.0067
p=0.2516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

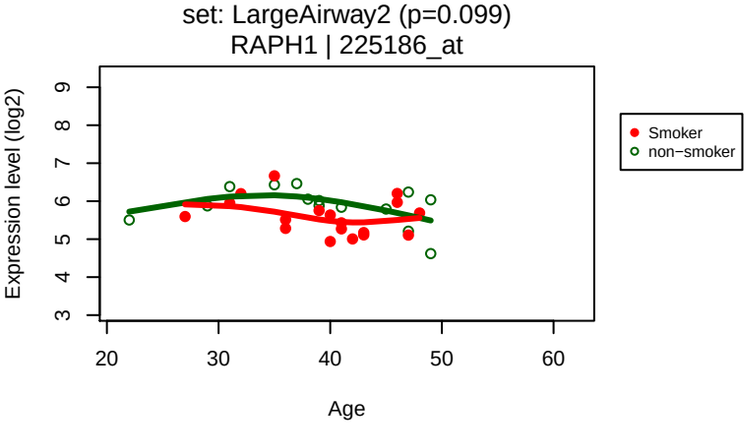

Gene expression ratio (smokers/nonsmokers)
gene: RAPH1 | 225186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.093
p=0
p=0
p=0.3194
p=0.099
p=0.7715
p=0.0067
p=0.2516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

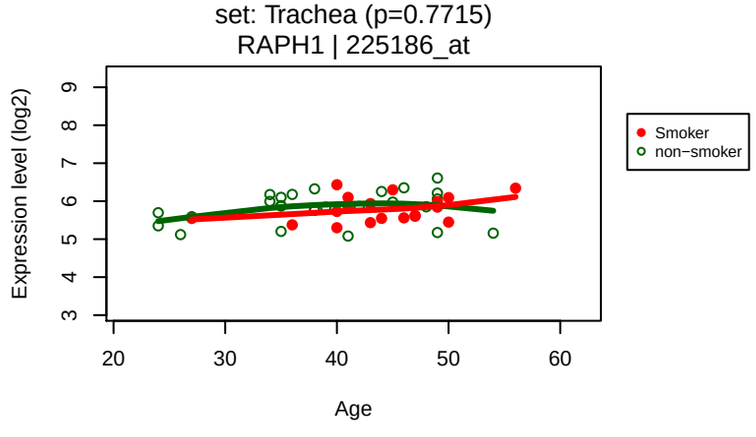

Gene expression ratio in Males
gene: RAPH1 | 225186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.2309
p=0.6347
p=0.3799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPH1 | 225186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0048
p=0.4157
p=0.9872
p=0.4677
SmallAirway2
LargeAirway2
Trachea
Macrophage2

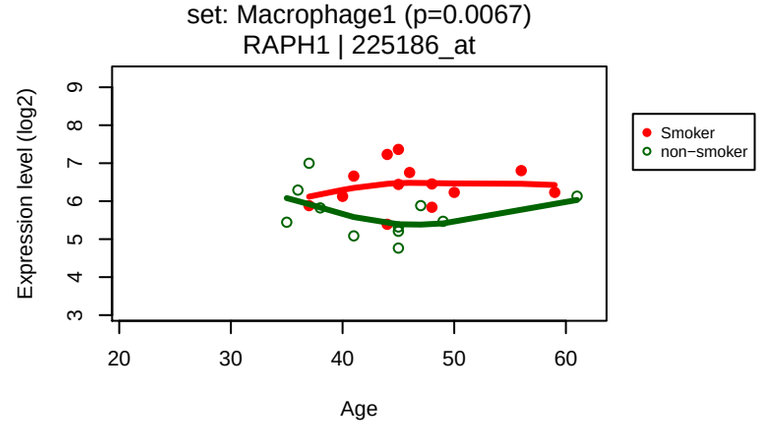

set: SmallAirway1 (p=0.093)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker

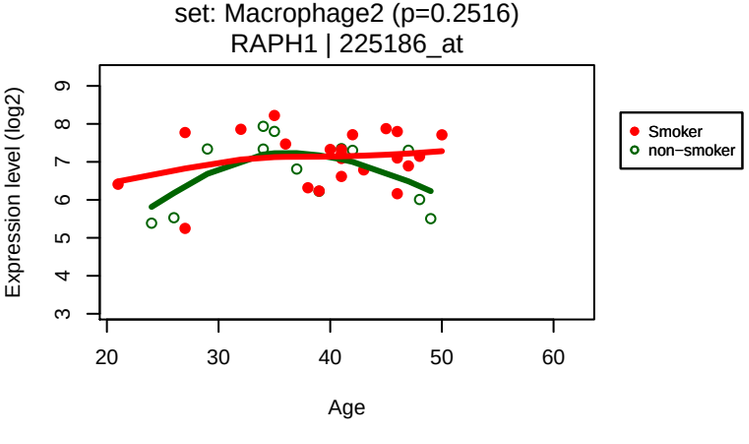

set: SmallAirway2 (p=0)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker

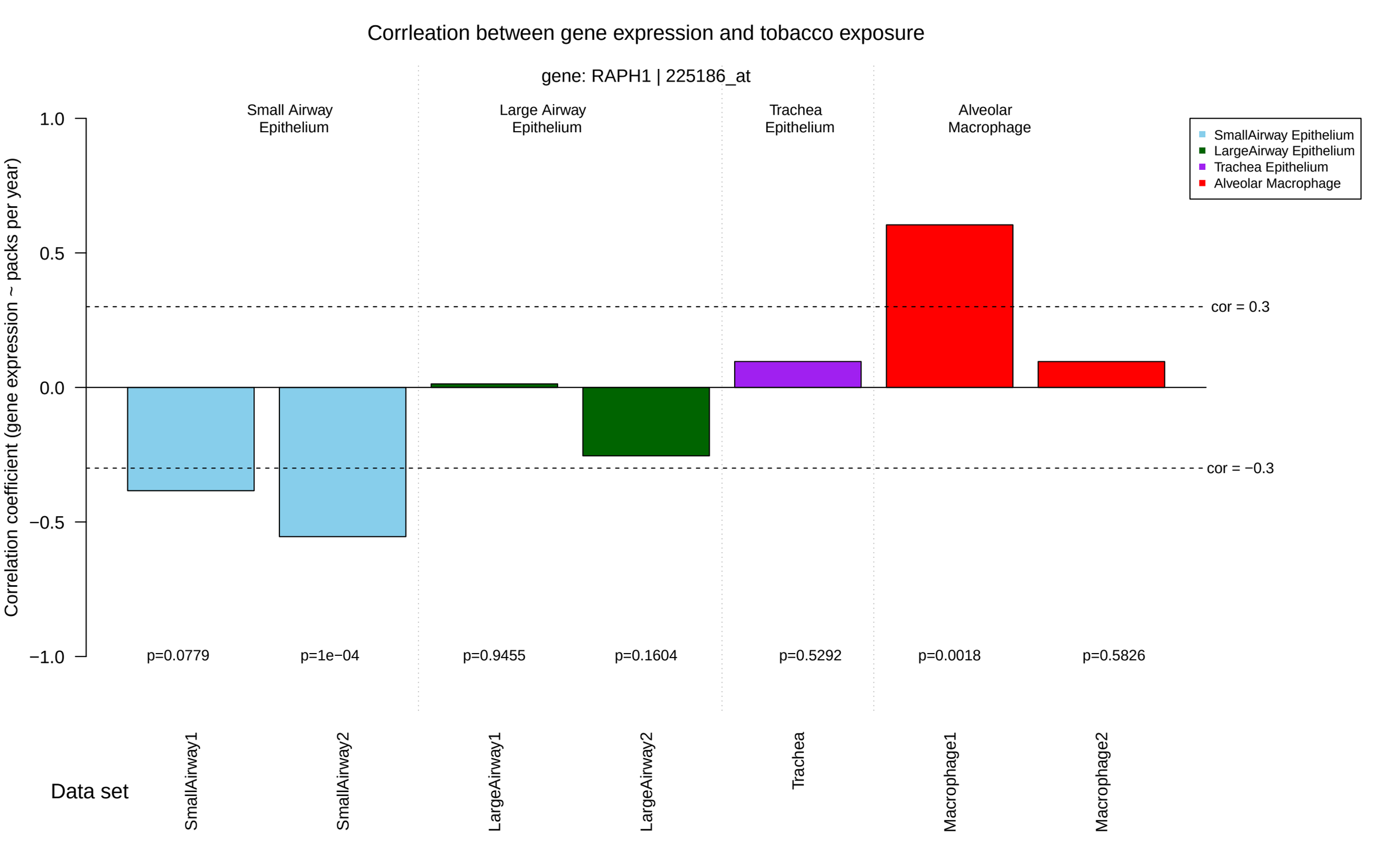

set: LargeAirway1 (p=0.3194)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.099)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7715)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0067)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2516)
RAPH1 | 225186_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPH1 | 225186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0779
p=1e−04
p=0.9455
p=0.1604
p=0.5292
p=0.0018
p=0.5826
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2